THE CARTESIAN COORDINATE SYSTEM

prepared by:

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GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

70%

WEEKLY

30%

EXAM

70%

LONG QUIZ

30%

SEATWORK

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

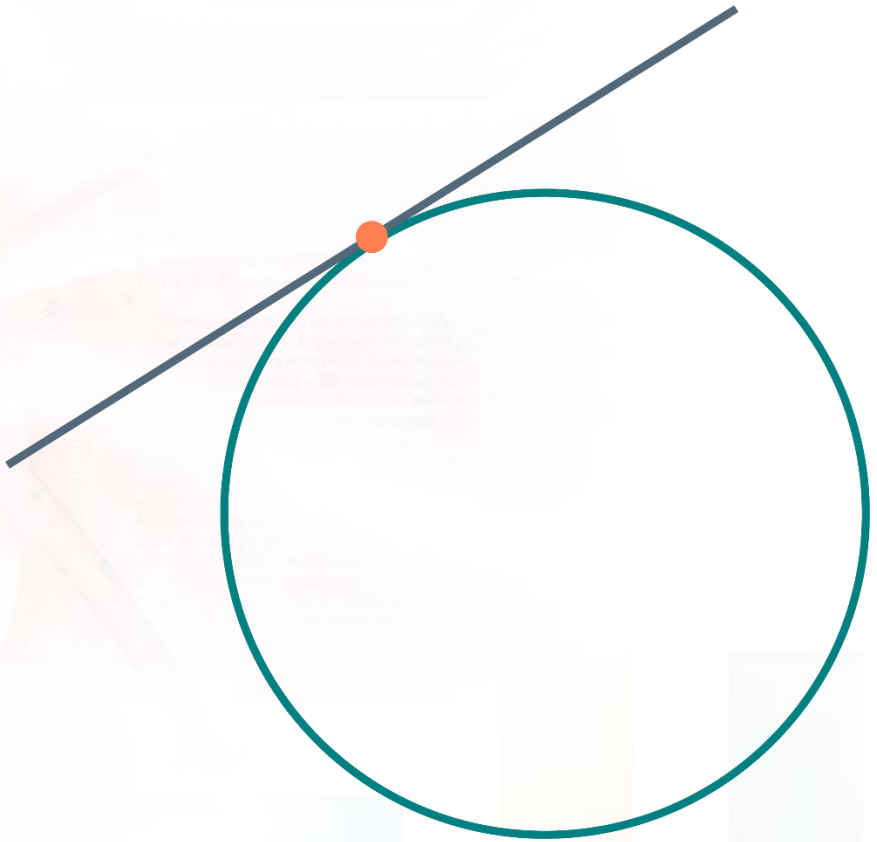
The Cartesian Coordinate System

- Rectangular Coordinates
- Polar Coordinates

MATH TELLS US 3 OF THE SADDEST LOVE STORIES

Tangent Line

who had **one chance** to meet and then departed forever

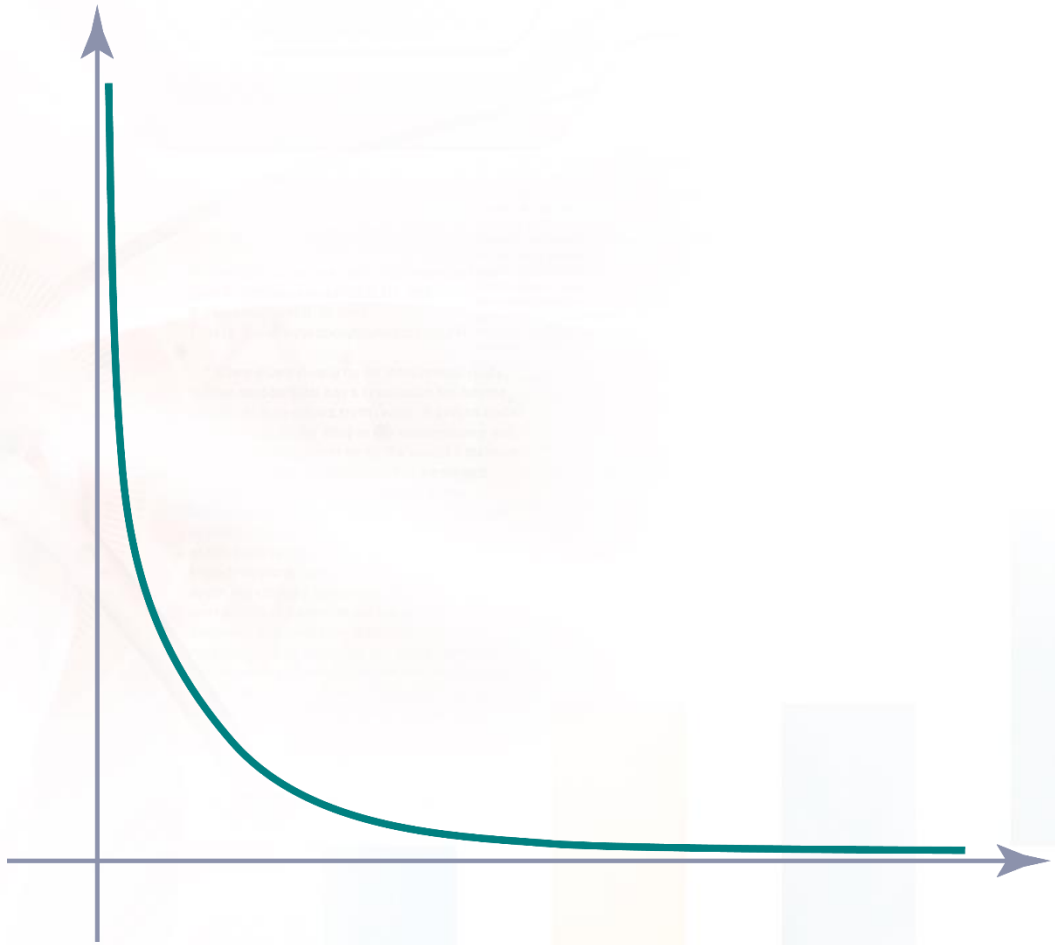


Parallel Line

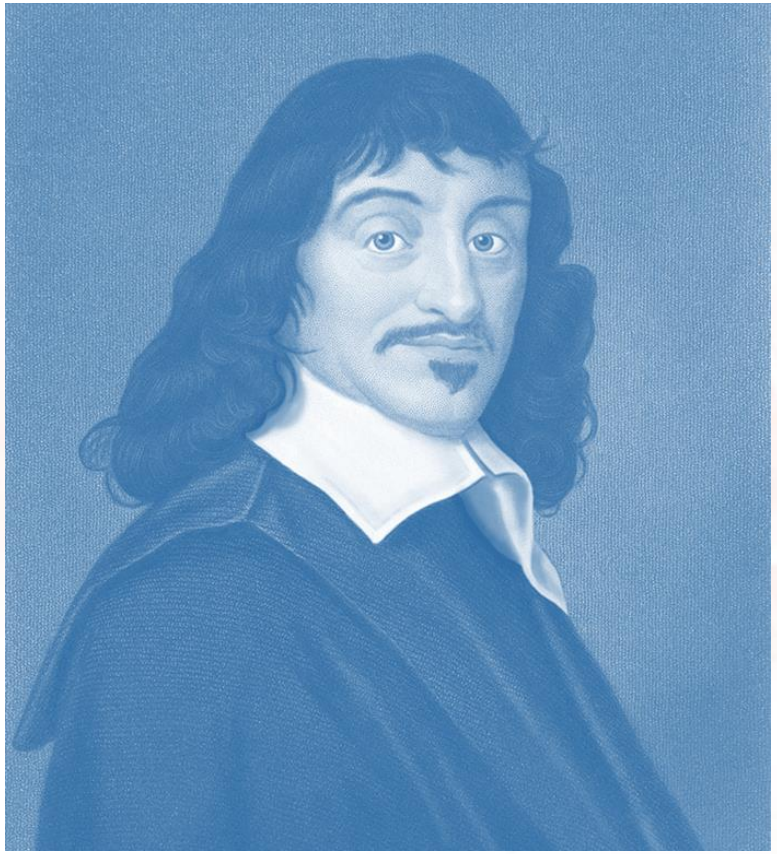
who were **never** meant to meet

Asymptote

who can get **closer and closer** to each other but will never be together



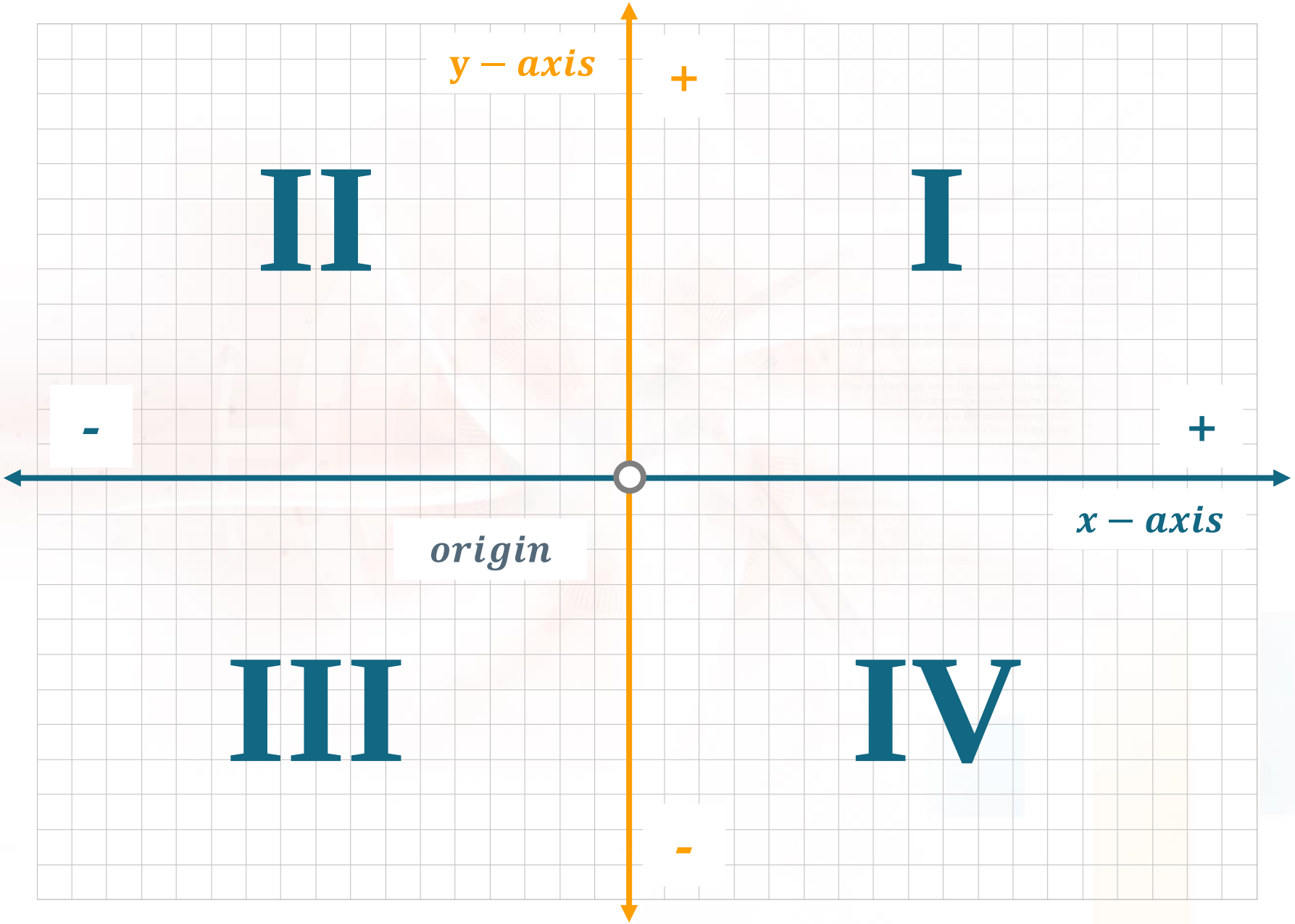
THE CARTESIAN COORDINATE SYSTEM



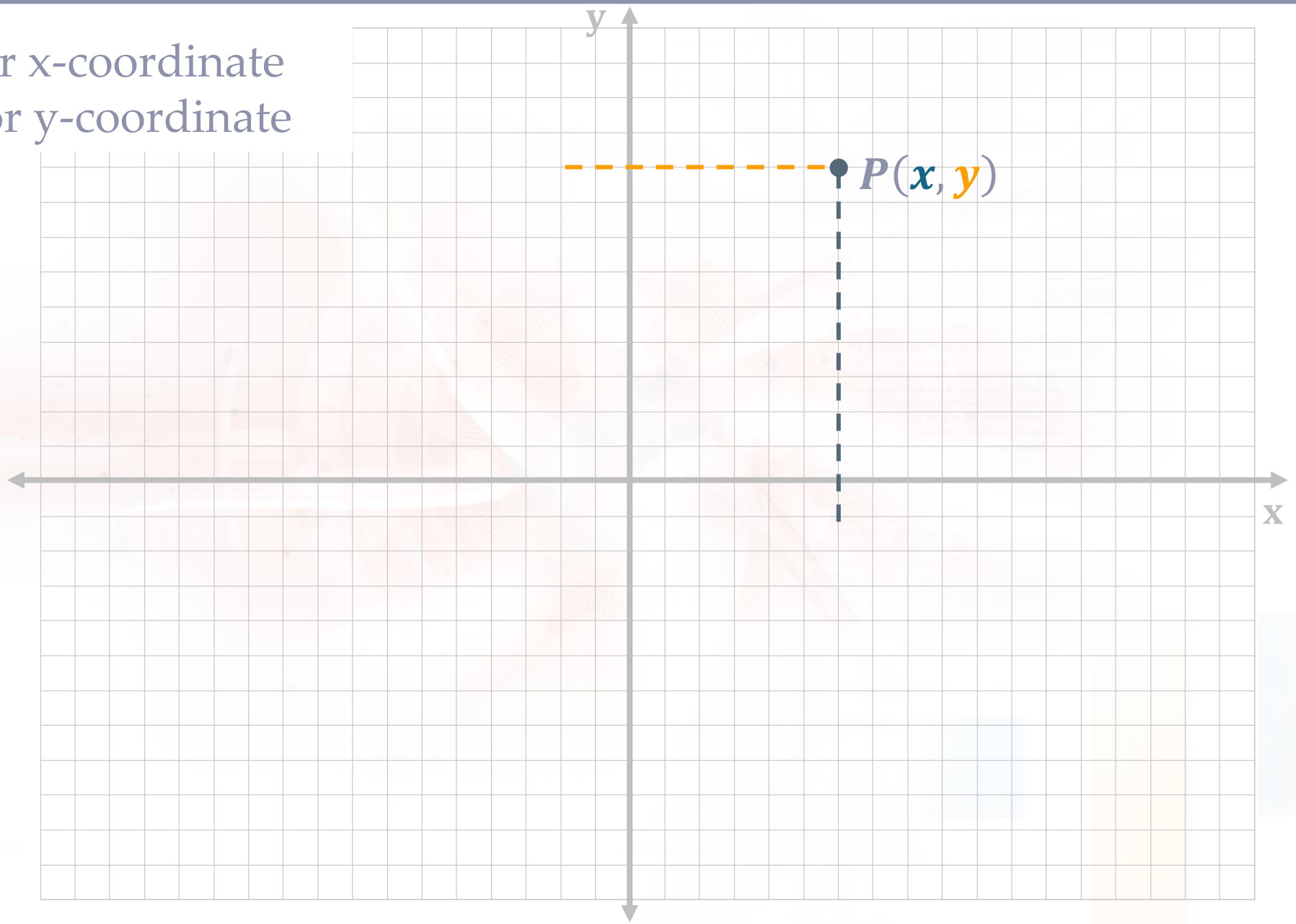
René Descartes

The story is that Descartes was lying in bed late one morning and saw a fly on the ceiling above him. As he watched the fly, he thought about how the **fly's location** could be described using the **edges** of the room for **reference**.

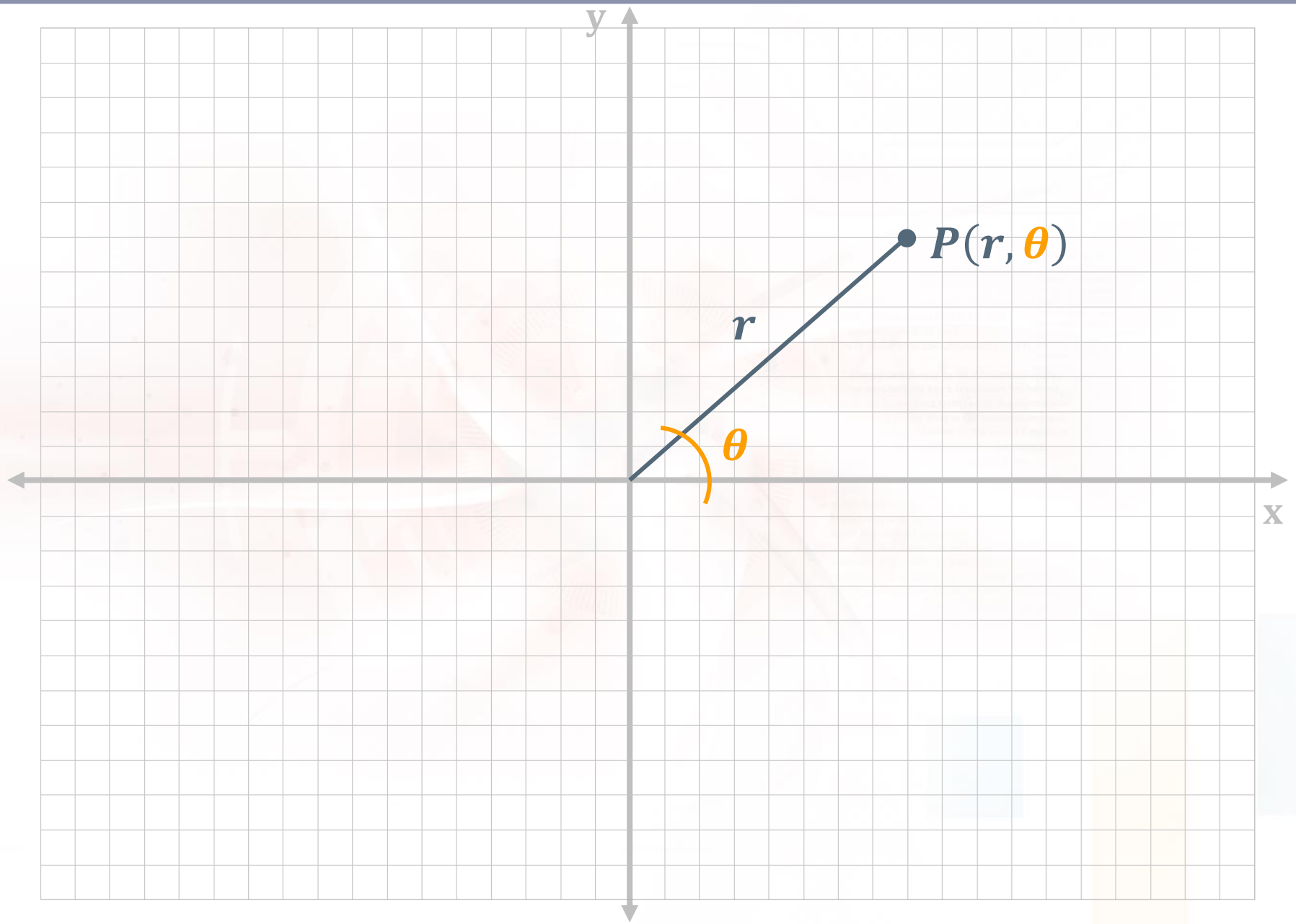
CARTESIAN PLANE



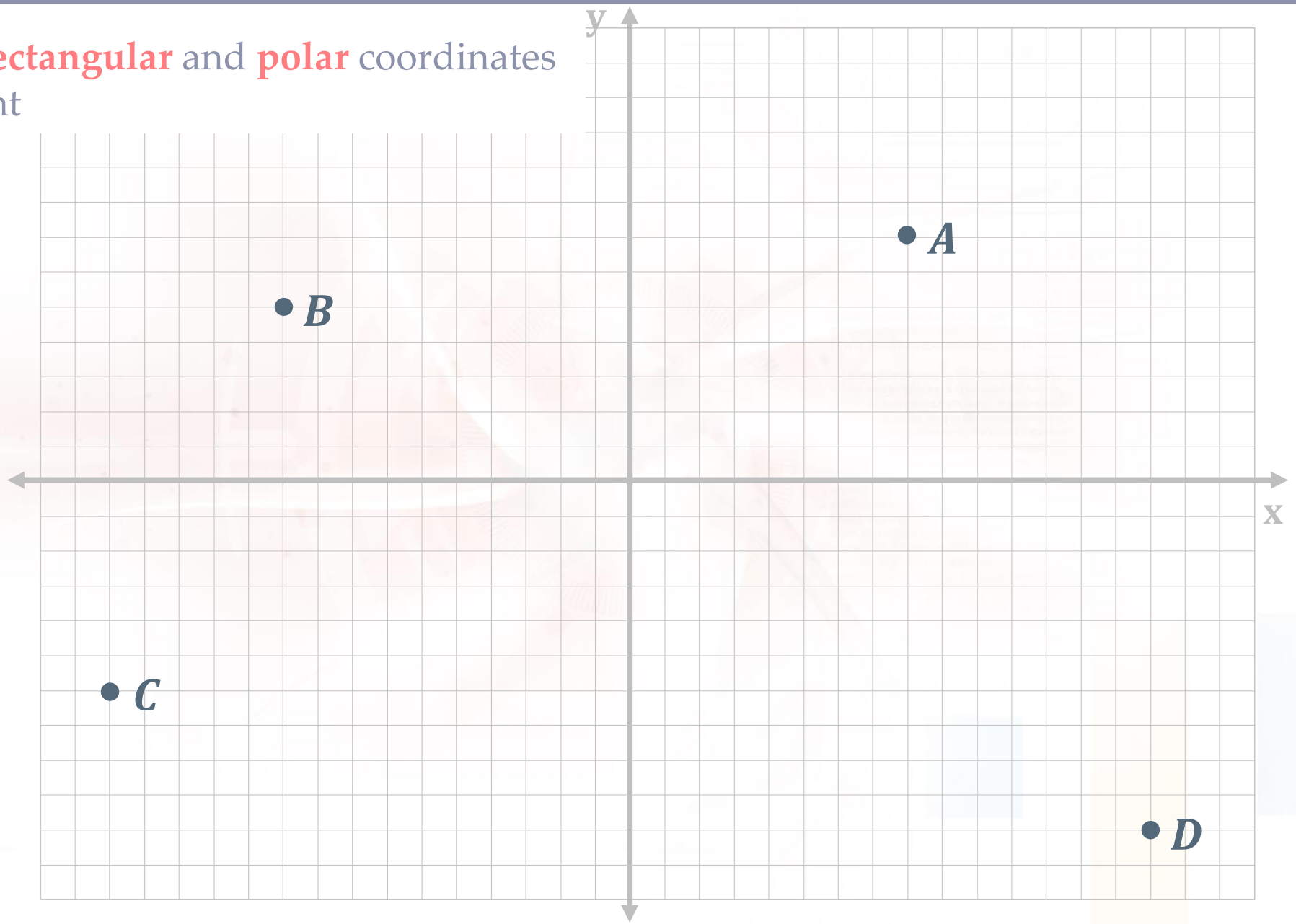
abscissa or x-coordinate
ordinate or y-coordinate



POLAR COORDINATES



Determine the **rectangular** and **polar** coordinates of the given point



POLAR ↔ RECTANGULAR

Given the point $P(r, \theta)$,

the **rectangular** coordinates are:

$$x = r \cos\theta$$

$$y = r \sin\theta$$

Given the point $P(x, y)$,

the **polar coordinates** are:

$$r^2 = x^2 + y^2$$

$$\tan\theta = \frac{y}{x}$$

Convert the polar coordinates $\boldsymbol{P\left(7,\frac{2\pi}{3}\right)}$ and

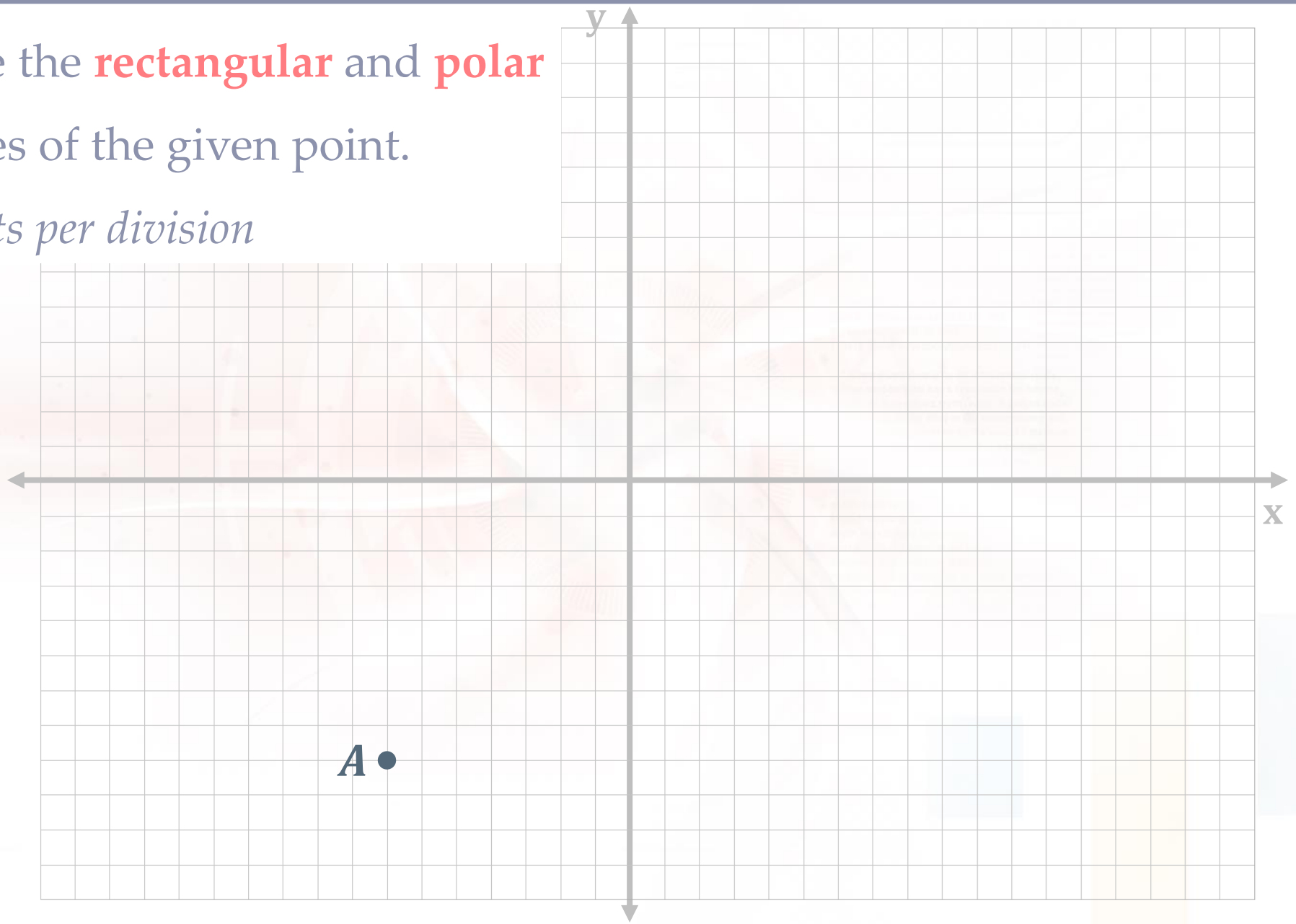
$\boldsymbol{P\left(19,\frac{5\pi}{4}\right)}$ to **rectangular** coordinates

Convert the rectangular coordinates $(\mathbf{13}, \mathbf{8})$ and $(\mathbf{-17}, \mathbf{22})$ to **polar** coordinates

SEATWORK

1. Determine the **rectangular** and **polar** coordinates of the given point.

   note: 2 units per division



2. Convert the polar coordinates $(\mathbf{13}, \mathbf{47°})$ to **rectangular** coordinates.

3. Convert the rectangular coordinates $(\mathbf{14}, \mathbf{-16})$ to **polar** coordinates.